



## Girraween High School

### 2015 Year 12 Trial Higher School Certificate

### Mathematics ( 2 unit )

#### General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen  
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

#### Section I

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

#### Section II

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

- For **Section II: Questions 11 – 16** MUST be returned in clearly marked *separate sections*.
- On each page of your answers, clearly write:
  - the **QUESTION** being answered
  - **YOUR NAME**
  - your **Mathematics TEACHER'S NAME**.
- Start each new question on a **NEW PAGE**.
- You may ask for extra pieces of paper if you need them.

**SECTION I****10 Marks****Attempt all of Questions 1 – 10****Allow about 15 minutes for this section.**

Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the value of  $\frac{2\pi^2}{19}$ , correct to 3 significant figures?

(A) 1.03  
(B) 1.04  
(C) 1.038  
(D) 1.039

2. Which of the following is equal to  $\frac{1}{3\sqrt{5} + \sqrt{2}}$ ?

(A)  $\frac{3\sqrt{5} - \sqrt{2}}{13}$   
(B)  $\frac{3\sqrt{5} + \sqrt{2}}{13}$   
(C)  $\frac{3\sqrt{5} - \sqrt{2}}{43}$   
(D)  $\frac{3\sqrt{5} + \sqrt{2}}{43}$

3. Which of the following is equivalent to  $\frac{\log_a 8}{\log_a 4}$ ?

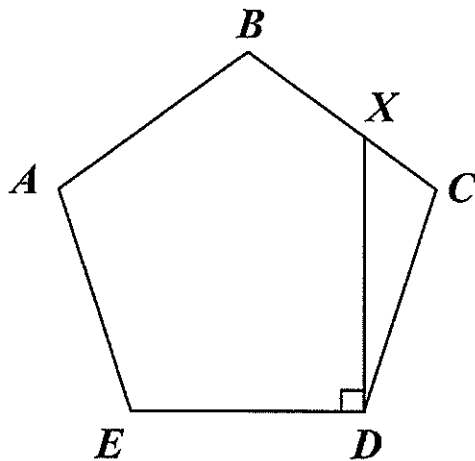
(A)  $1\frac{1}{2}$   
(B) 2  
(C)  $\log_a 2$   
(D)  $\log_a 4$

4. The quadratic equation  $2x^2 - 4x - 3 = 0$  has roots  $\alpha$  and  $\beta$ .

What is the value of  $(\alpha + \beta) - (\alpha\beta)$ ?

- (A)  $-\frac{1}{2}$   
 (B)  $-3\frac{1}{2}$   
 (C)  $\frac{1}{2}$   
 (D)  $3\frac{1}{2}$

5.



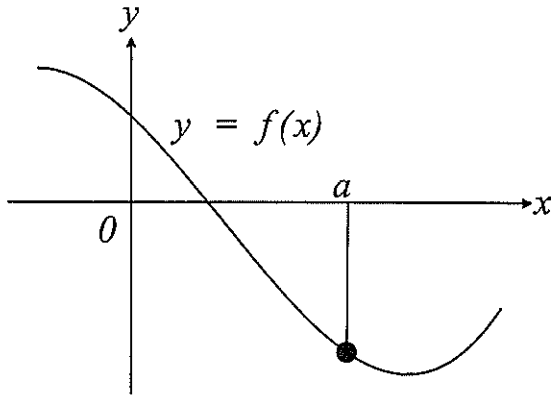
NOT TO SCALE

$ABCDE$  is a regular pentagon and  $DX \perp DE$ .

The size of  $\angle CXD$  is :

- (A)  $36^\circ$   
 (B)  $54^\circ$   
 (C)  $64^\circ$   
 (D)  $72^\circ$

6. The diagram shows the graph of  $y = f(x)$ .



Which of the following statements is true?

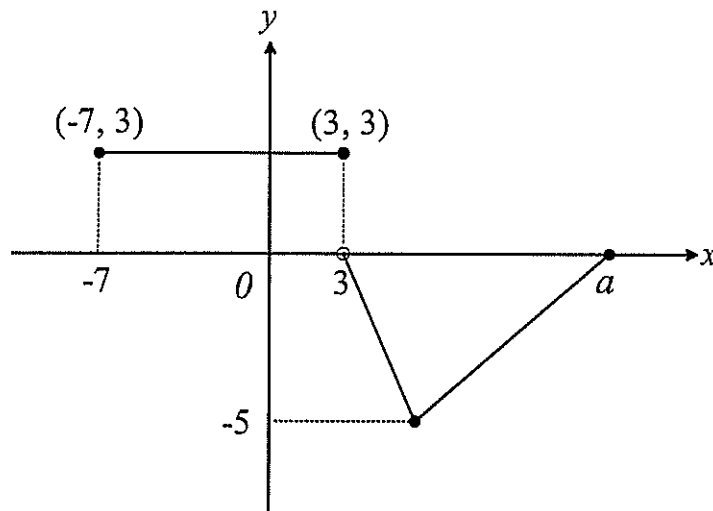
- (A)  $f'(a) < 0$  and  $f''(a) < 0$   
 (B)  $f'(a) < 0$  and  $f''(a) > 0$   
 (C)  $f'(a) > 0$  and  $f''(a) < 0$   
 (D)  $f'(a) > 0$  and  $f''(a) > 0$
7. A parabola has focus  $(0, -4)$  and directrix  $y = 2$ .

What is the equation of the parabola?

- (A)  $x^2 = -12(y + 1)$   
 (B)  $x^2 = -24(y + 4)$   
 (C)  $x^2 = 12(y + 1)$   
 (D)  $x^2 = 24(y + 4)$
8. What is the derivative of  $\frac{x}{\sin x}$ ?

- (A)  $\frac{-x \cos x - \sin x}{\sin^2 x}$   
 (B)  $\frac{x \cos x - \sin x}{\sin^2 x}$   
 (C)  $\frac{\sin x + x \cos x}{\sin^2 x}$   
 (D)  $\frac{\sin x - x \cos x}{\sin^2 x}$

9. The diagram shows the graph of  $y = f(x)$ .

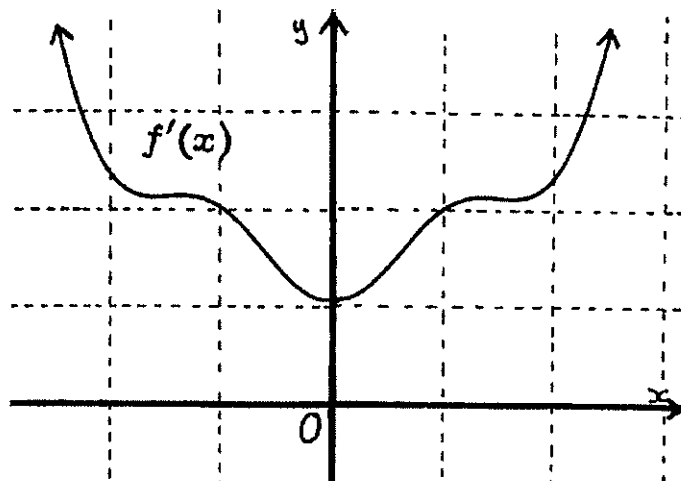


NOT TO SCALE

Use the graph to determine the value of  $a$  which satisfies the condition  $\int_{-7}^a f(x) \, dx = 0$ .

- (A) 9  
(B) 12  
(C) 13  
(D) 15

10.



It is known that  $y = f(x)$  passes through the origin.

By examining the graph of  $y = f'(x)$  shown above, state which of the following statements is true?

- (A)  $f(x)$  is an even function with a point of inflexion at  $f(0)$ .  
(B)  $f(x)$  is an even function with a local maximum at  $f(0)$ .  
(C)  $f(x)$  is an odd function with a point of inflexion at  $f(0)$ .  
(D)  $f(x)$  is an odd function with a local maximum at  $f(0)$ .

The examination continues on the next page.

**SECTION II****90 Marks****Attempt all of Questions 11 – 16****Allow about 2 hours and 45 minutes for this section.**

Answer each question on the paper provided.

Start each new question on a **NEW PAGE**.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

**Question 11 (15 Marks)****Marks**Start a **NEW PAGE**.

- (a) Factorise  $2x^2 + 11x - 21$ . 2
- (b) Solve  $|3x - 5| < 4$ . 2
- (c) Find the equation of the tangent to the curve  $y = x^3$  at the point where  $x = 2$ . 2
- (d) Differentiate  $(e^{3x} - 5)^4$ . 2
- (e) Evaluate  $\int_1^2 \frac{1}{(3x - 1)^2} dx$  3
- (f) (i) Find  $\frac{d}{dx}(\sin x^3)$ . 2
- (ii) Hence find  $\int x^2 \cos x^3 dx$ . 2

**The examination continues on the next page.**

**Question 12 (15 Marks)****Marks**

Start a NEW PAGE.

- (a) What is the radius of a circle in which an arc of length 15 cm subtends an angle of  $60^\circ$  at the centre? 2

Give your answer correct to the nearest mm.

- (b) Solve  $2 \log_e x = \log_e (3x + 10)$ . 2

- (c) Differentiate  $y = \frac{\sin x}{1 + \cos x}$  3

and hence show that  $\frac{dy}{dx} = \frac{1}{1 + \cos x}$ .

- (d) Find  $\int \frac{10x}{x^2 - 8} dx$ . 2

- (e) Evaluate  $\int_0^{\frac{\pi}{4}} \cos(2x) dx$  2

- (f) Mayomi and Christina each throw a die.

- (i) Find the probability that they throw the same number. 2

- (ii) Find the probability that the number thrown by Christina is greater than the number thrown by Mayomi. 2

**The examination continues on the next page.**

**Question 13 (15 Marks)**

**Marks**

Start a NEW PAGE.

- (a) The first and last terms of an arithmetic series are 10 and 60 respectively, and the sum of the series is 3535.

Find:

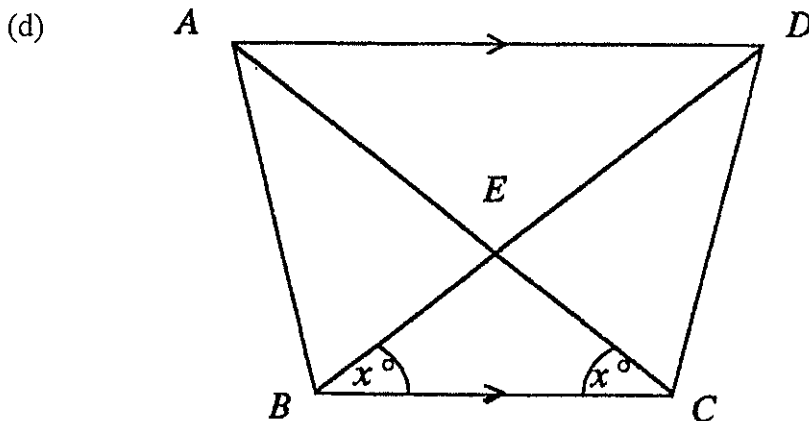
- (i) the number of terms in the series. 2  
 (ii) the common difference. 2

- (b) “Although the number of unemployed is increasing, the Government’s policies to reduce unemployment seem to be taking effect.” 2

Given  $E$  is the number of unemployed, what does the above statement imply about

$$\frac{dE}{dt} \text{ and } \frac{d^2 E}{dt^2} ?$$

- (c) What is the volume of the solid of revolution formed by rotating the curve  $y = \sec x$  about the  $x$ -axis for  $0 \leq x \leq \frac{\pi}{3}$ ? 2



In the diagram  $AD$  is parallel to  $BC$  and  $\angle DBC = \angle ACB = x^\circ$ .

- (i) Show that  $AE = DE$ . 2  
 (ii) Prove that the triangles  $\triangle ABC$  and  $\triangle DCB$  are congruent. 3  
 (iii) Deduce that  $\angle ABD = \angle DCA$ . 2

**The examination continues on the next page.**

**Question 14 (15 Marks)****Marks**

Start a NEW PAGE.

(a) Consider the geometric series  $1 + (\sqrt{5} - 2) + (\sqrt{5} - 2)^2 + \dots$

(i) Explain why the geometric series has a limiting sum.

**1**

(ii) Find the exact value of the limiting sum.

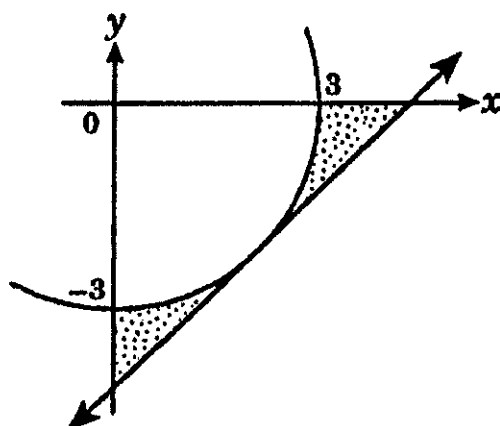
**2**

Write your answer with a rational denominator.

(b) Find the values of  $k$  for which the quadratic equation  $x^2 + (k - 2)x + 4 = 0$  has no real roots.

**2**

(c) (i) Prove that the line  $3x - 4y = 15$  is a tangent to the circle  $x^2 + y^2 = 9$ .

**2**

(ii) Calculate the exact area in the fourth quadrant between the tangent  $3x - 4y = 15$  and the circle  $x^2 + y^2 = 9$  (as shown by the shaded area in the diagram above).

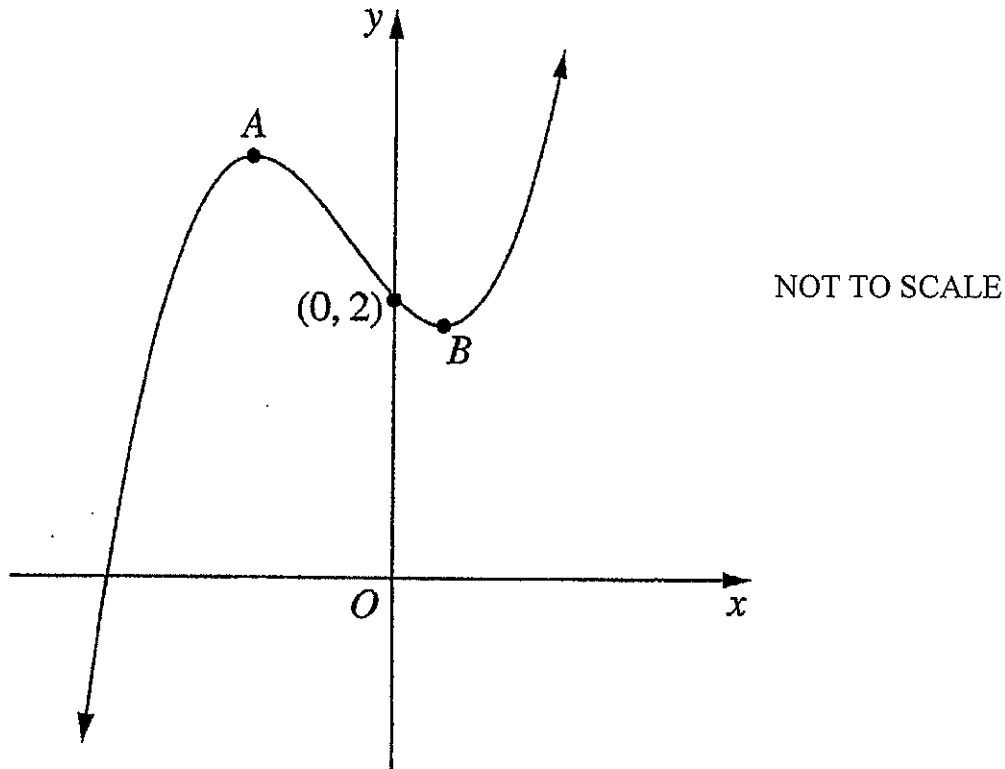
**2**

**Question 14 continues on the next page.**

**Question 14 (continued)**

- (d) The graph of  $y = x^3 + x^2 - x + 2$  is sketched below.

The points  $A$  and  $B$  are the turning points.



- |       |                                                                                       |   |
|-------|---------------------------------------------------------------------------------------|---|
| (i)   | Find the coordinates of $A$ and $B$ .                                                 | 3 |
| (ii)  | For what values of $x$ is the curve concave up?<br>Give reasons for your answer.      | 2 |
| (iii) | For what values of $k$ has the equation $x^3 + x^2 - x + 2 = k$ three real solutions? | 1 |

**End of Question 14.**

**The examination continues on the next page.**

**Question 15 (15 Marks)**

**Marks**

Start a NEW PAGE.

- (a) A standard pack of 52 cards consists of four suits (Diamonds, Hearts, Clubs and Spades) with 13 cards in each suit.

- (i) One card is drawn from the pack and kept on the table. A second card is drawn and placed beside it on the table. 1

What is the probability that the second card is from a different suit to the first?

- (ii) The two cards are replaced and the pack shuffled. Four cards are chosen from the pack and placed side by side on the table. 2

What is the probability that these four cards are all from different suits?

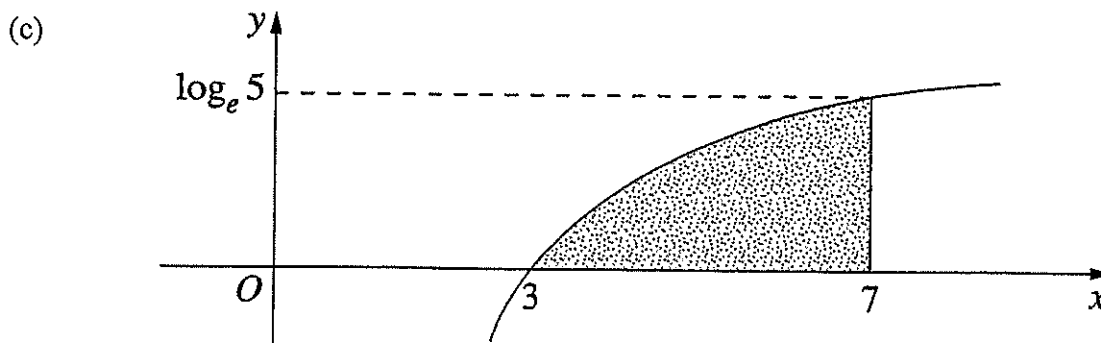
- (b) A particle is initially at rest at the origin.

Its acceleration as a function of time  $t$  is given by  $\ddot{x} = 4 \sin 2t$ .

- (i) Show that the velocity of the particle is given by  $\dot{x} = 2 - 2 \cos 2t$ . 2

- (ii) Sketch the graph of the velocity for  $0 \leq t \leq 2\pi$  AND determine the time at which the particle first comes to rest after  $t = 0$ . 3

- (iii) Find the distance travelled by the particle between  $t = 0$  and the time at which the particle first comes to rest after  $t = 0$ . 2



In the diagram, the shaded region is bounded by  $y = \log_e(x - 2)$ , the  $x$ -axis and the line  $x = 7$ . 5

Find the exact value of the area of the shaded region.

**The examination continues on the next page.**

**Question 16 (15 Marks)****Marks**

Start a NEW PAGE.

- (a) A drug is used to control a medical condition. It is known that the quantity  $Q$  of drug remaining in the body after  $t$  hours satisfies an equation of the form

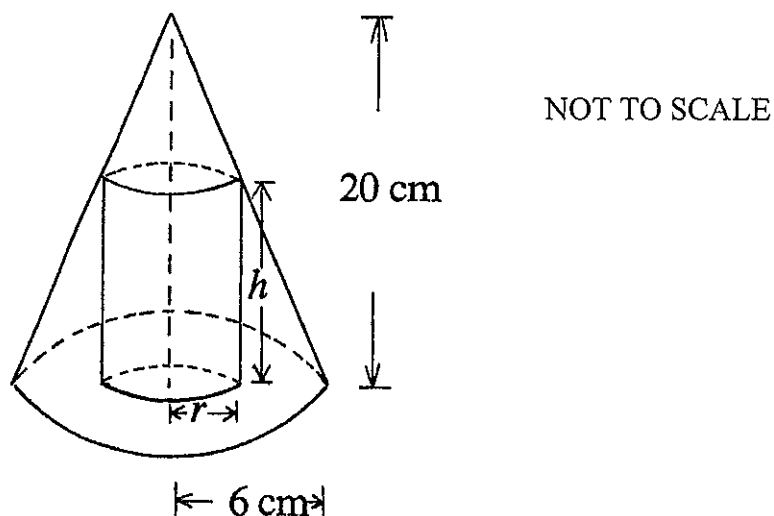
$$Q = Q_0 e^{-kt}$$

where  $Q_0$  and  $k$  are constants.

The initial dose is 6 milligrams and after 15 hours the amount remaining in the body is half the initial dose.

- (i) Find the values of  $Q_0$  and  $k$ , correct to 3 decimal places where necessary. 2
- (ii) When will one-eighth of the initial dose remain? 2

(b)



A cylinder of radius  $r$  cm and height  $h$  cm is inscribed in a cone with base radius 6 cm and height 20 cm as in the diagram.

- (i) Show that the volume  $V$  of the cylinder is given by 2

$$V = \frac{10 \pi r^2 (6 - r)}{3}.$$

- (ii) Hence find the values of  $r$  and  $h$  for the cylinder which has maximum volume. 2

**Question 16 continues on the next page.**

**Question 16 (continued)**

- (c) A farmer borrows \$80 000 to purchase new machinery. The interest is calculated monthly at the rate of 2% per month, and is compounded each month.

The farmer intends to repay the loan with interest in two equal annual instalments of \$ $M$  at the end of the first and second years.

- (i) How much does the farmer owe at the end of the first month? 1
- (ii) Write an expression involving  $M$  for the total amount owed by the farmer after 12 months, just after the first instalment of \$ $M$  has been paid. 2
- (iii) Find an expression for the amount owed at the end of the second year and deduce that 2

$$M = \frac{80\,000 \times (1.02)^{24}}{(1.02)^{12} + 1}$$

- (iv) What is the total interest over the two year period? 2

**End of examination.**

# MULTIPLE-CHOICE

- 1) B 2) C 3) A 4) D 5) B  
6) B 7) A 8) D 9) D 10) C

$$1) \frac{2\pi^2}{19} = 1.038905726$$

$$= 1.04 \text{ (3 sig figs)} \quad \textcircled{B}$$

$$2) \frac{1}{3\sqrt{5}+\sqrt{2}} \times \frac{3\sqrt{5}-\sqrt{2}}{3\sqrt{5}-\sqrt{2}} = \frac{3\sqrt{5}-\sqrt{2}}{9 \times 5 - 2}$$

$$= \frac{3\sqrt{5}-\sqrt{2}}{43} \quad \textcircled{C}$$

$$3) \frac{\log_a 8}{\log_a 4} = \frac{\log_a 2^3}{\log_a 2^2}$$

$$= \frac{3 \log_a 2}{2 \log_a 2}$$

$$= \frac{3}{2} \text{ or } 1\frac{1}{2} \quad \textcircled{A}$$

$$4) 2x^2 - 4x - 3 = 0 \quad \alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

$$= -(-4) = 4 \quad = -\frac{3}{2}$$

$$(\alpha + \beta) - (\alpha\beta) = 4 - (-\frac{3}{2})$$

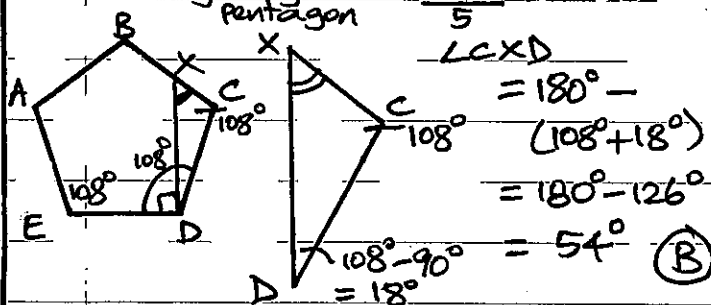
$$= 4 + \frac{3}{2} = 5\frac{1}{2} \quad \textcircled{D}$$

$$5) \text{Angle sum pentagon} = (n-2) \times 180^\circ$$

$$(5 \text{ sided}) = 3 \times 180^\circ$$

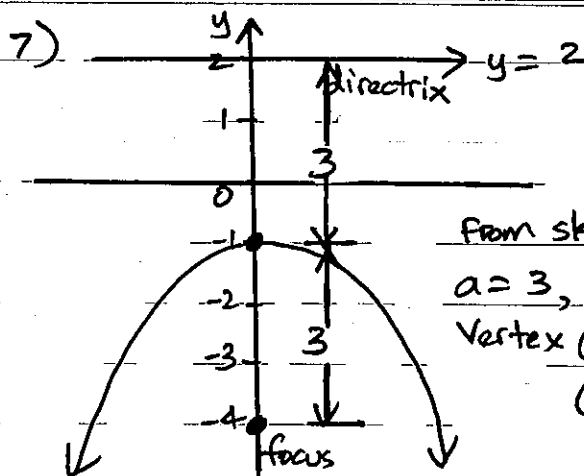
$$= 540^\circ$$

$$\therefore \text{each angle regular} = \frac{540^\circ}{5} = 108^\circ$$



6)  $\frac{dy}{dx} < 0$   $\frac{d^2y}{dx^2} > 0$

At  $x=a$ :  $f'(a) < 0$  and  $f''(a) > 0$   $\textcircled{B}$



Parabola of form  $(x-h)^2 = -4a(y-k)$

$$(x-0)^2 = -4 \times 3(y-(-1))$$

$$x^2 = -12(y+1) \quad \textcircled{A}$$

$$8) y = \frac{x}{\sin x} \quad u = x \quad v = \sin x$$

$$\frac{du}{dx} = 1 \quad \frac{dv}{dx} = \cos x$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(\sin x)(1) - (x) \cos x}{\sin^2 x}$$

$$= \frac{\sin x - x \cos x}{\sin^2 x} \quad \textcircled{D}$$

9)  $A_1 = 3 \times 10 = 30 \text{ units}^2$

For  $\int_{-7}^a f(x) dx = 0$

Require  $A_1 = A_2$

$$A_2 = \frac{1}{2}bh = \frac{1}{2} \times b \times 5 = 30$$

$$5b = 60$$

$$b = 12$$

$$\therefore a = 12 + 3 = 15 \quad \textcircled{D}$$

10) From graph of  $y = f'(x)$ :

$f'(x) > 0$  for all  $x$

So  $f(x)$  ALWAYS increasing

$\therefore$  No stationary points/No Max/min turning points.

at  $x=0$ :  $f'(x)$  has stationary point

$\therefore$  At  $x=0$ ,  $f''(x)=0$  i.e.  $f''(0)=0$

So  $x=0$  may be point of inflexion.

Need to check for change of concavity "around"  $x=0$ :

- to immediate left of  $x=0$ :  $f''(x) < 0$  concave DOWN
- to immediate right of  $x=0$ :  $f''(x) > 0$  concave UP

$\therefore$  at  $x=0$ ,  $f''(0)=0$  & concavity changes  $\rightarrow$  point of inflexion.

ODD function:  $f(x)$  will be symmetrical about the origin

$\therefore$  ODD function, pt inflexion  $x=0$   $\textcircled{C}$

Question 11 (15 Marks)

$$\begin{aligned}
 \text{(a)} \quad & 2x^2 + 11x - 21 \quad \begin{array}{l} -42 \quad -42 \quad \frac{14}{-3} \\ -11 \end{array} \\
 & = 2x^2 + 14x - 3x - 21 \\
 & = 2x(x+7) - 3(x+7) \\
 & = (2x-3)(x+7) \quad (2m)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & |3x-5| < 4 \\
 & \Rightarrow (3x-5) < 4 \quad \text{or} \quad -(3x-5) < 4 \\
 & 3x < 9 \quad \quad \quad 3x-5 > -4 \\
 & x < 3 \quad \quad \quad 3x > 1 \\
 & \quad \quad \quad x > \frac{1}{3} \\
 & \therefore \frac{1}{3} < x < 3 \quad (2m)
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad & y = x^3 \\
 & \frac{dy}{dx} = 3x^2 \\
 \text{when } x=2: & \frac{dy}{dx} = 3 \times 4 \\
 & \frac{dy}{dx} = 12 \\
 \text{when } x=2: & y = 2^3 \\
 & y = 8 \\
 \text{Require tangent } m=12, (2, 8): \\
 & y - y_1 = m(x - x_1) \\
 & y - 8 = 12(x - 2) \\
 & y - 8 = 12x - 24 \\
 & y = 12x - 16 \\
 \text{or } 12x - y - 16 = 0 \quad (2m)
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad & y = (e^{3x} - 5)^4 \\
 & \frac{dy}{dx} = 4(e^{3x} - 5)^3 \cdot (3e^{3x}) \\
 & = 12e^{3x} (e^{3x} - 5)^4 \quad (2m)
 \end{aligned}$$

$$\begin{aligned}
 \text{(e)} \quad & \int_1^2 \frac{1}{(3x-1)^2} dx = \int_1^2 (3x-1)^{-2} dx \\
 & = \left[ \frac{(3x-1)^{-1}}{3(-1)} \right]_1^2 \\
 & = \left[ \frac{-1}{3(3x-1)} \right]_1^2 \\
 & = \frac{-1}{15} - \left( \frac{-1}{6} \right) \\
 & = \frac{-1}{15} + \frac{1}{6} \\
 & = \frac{1}{10} \quad (3m)
 \end{aligned}$$

Question 11 continued:

$$\begin{aligned}
 \text{11) (f) (i)} \quad & y = \sin x^3 \\
 & y = \sin u \quad u = x^3 \\
 & \frac{dy}{du} = \cos u \quad \frac{du}{dx} = 3x^2 \\
 & \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \\
 & = \cos u \times 3x^2 \\
 & = 3x^2 \cos x^3 \quad (2m)
 \end{aligned}$$

$$\begin{aligned}
 \text{f) (ii)} \quad & \text{If } \frac{d}{dx} (\sin x^3) = 3x^2 \cos x^3 \\
 & \text{then } \int 3x^2 \cos x^3 dx = \sin x^3 + C_1 \\
 & \therefore \int x^2 \cos x^3 dx = \frac{1}{3} \sin x^3 + C \quad (2m)
 \end{aligned}$$

Question 12 (15 Marks)

$$\begin{aligned}
 \text{(a)} \quad & L = 15 \quad \theta = 60^\circ \\
 & \quad \quad \quad = \frac{\pi}{3} \\
 & l = r\theta \\
 & 15 = r \cdot \frac{\pi}{3} \\
 & r = \frac{45}{\pi} \\
 & r = 14.32394488 \\
 & \therefore r = 14.3 \text{ cm, or } r = 143 \text{ mm} \\
 & \text{(correct to nearest mm.)} \quad (2m)
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & 2 \log_e x = \log_e (3x+10) \\
 & \log_e x^2 = \log_e (3x+10) \\
 & \therefore x^2 = 3x+10 \\
 & x^2 - 3x - 10 = 0 \\
 & (x+2)(x-5) = 0 \\
 & x = -2, \text{ or } x = 5 \\
 & x \neq -2 \text{ as for } y = \log_e x, x > 0. \\
 & \therefore \text{only solution is } x = 5. \quad (2m)
 \end{aligned}$$

Question 12 continued:

$$12)(c) \quad y = \frac{\sin x}{1 + \cos x}$$

$$u = \sin x \quad v = 1 + \cos x$$

$$\frac{du}{dx} = \cos x \quad \frac{dv}{dx} = -\sin x$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(1 + \cos x)(\cos x) - (\sin x)(-\sin x)}{(1 + \cos x)^2}$$

$$= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2}$$

$$= \frac{\cos x + 1}{(1 + \cos x)^2} \quad \text{since } \sin^2 x + \cos^2 x = 1$$

$$\therefore \frac{dy}{dx} = \frac{1}{1 + \cos x}$$

(3m)

$$12)(d) \quad \int \frac{10x}{x^2-8} dx = 5 \int \frac{2x}{x^2-8} dx$$

$$= 5 \log_e(x^2-8) + C$$

$$\text{since } \int \frac{f'(x)}{f(x)} dx = \log_e f(x) + C$$

(2m)

$$12)(e) \quad \int_0^{\frac{\pi}{4}} \cos(2x) dx$$

$$= \frac{1}{2} \left[ \sin(2x) \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left[ \sin \frac{\pi}{2} - \sin 0 \right]$$

$$= \frac{1}{2} (1 - 0)$$

$$= \frac{1}{2}$$

$$12)(f)(i) \quad P(\text{throw same number})$$

$$= P(1,1) \text{ or } P(2,2) \text{ or } P(3,3) \text{ or } P(4,4) \\ \text{or } P(5,5) \text{ or } P(6,6)$$

$$= 6 \times \frac{1}{36}$$

$$= \frac{6}{36}$$

$$= \frac{1}{6}$$

(2m)

Question 12(f) continued:

12)(f)(ii) Let Mayomi be first number, then Christina throw second number.  
Possible outcomes:

(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)  
(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)  
(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)  
(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)  
(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)  
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)

15 possible outcomes where Christina throws number greater than Mayomi.

$$\therefore P(\text{Christina's number} > \text{Mayomi's number})$$

$$= \frac{15}{36}$$

$$= \frac{5}{12}$$

(2m)

12)(f)(ii) ALTERNATIVE METHOD

from f(i),  $P(\text{throw same number}) = \frac{1}{6}$

$\therefore P(\text{do NOT throw same number}) = \frac{5}{6}$

of this,  $\frac{1}{2} \times \frac{5}{6} = P(\text{Christina} > \text{Mayomi})$   
 $= P(\text{Mayomi} > \text{Christina})$

$\therefore P(\text{Christina's number} > \text{Mayomi's number})$

$$= \frac{1}{2} \times \frac{5}{6}$$

$$= \frac{5}{12}$$

(2m)

Question 13 (15 Marks)

(a)(i) Arithmetic series:  $a = 10, l = 60$   
 $S_n = 3535$

$$S_n = \frac{n}{2} (a + l)$$

$$3535 = \frac{n}{2} (10 + 60)$$

$$35n = 3535$$

$$n = \frac{3535}{35}$$

$$\therefore n = 101$$

(2m)

(b)(a)(ii)  $l = a + (n-1)d$

$$60 = 10 + 100d$$

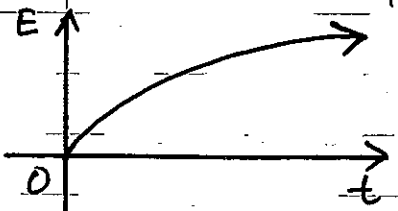
$$100d = 50$$

$$\therefore d = \frac{1}{2}$$

(2m)

Question 13 continued:

(13)(b) Possible graph of unemployment <sup>(E)</sup> over time may be:

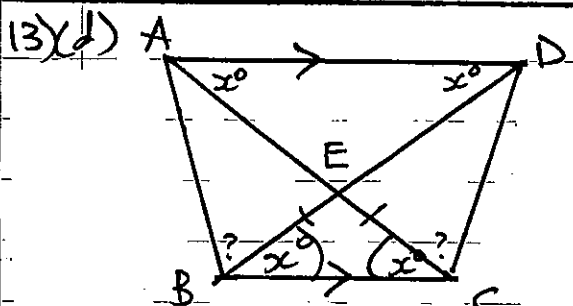


Unemployment is increasing: so  $\frac{dE}{dt} > 0$ .  
 Since policy to reduce unemployment is taking effect, then even though the number of unemployed is still increasing, rate of increase is slowing down, i.e. curve is concave down.

i.e.  $\frac{d^2E}{dt^2} < 0$

so in this case:  $\frac{dE}{dt} > 0$ ,  $\frac{d^2E}{dt^2} < 0$ . (2m)

(13)(c) volume about x-axis =  $\pi \int y^2 dx$   
 $= \pi \int_0^{\frac{\pi}{3}} \sec^2 x dx$   
 $= \pi [\tan x]_0^{\frac{\pi}{3}}$   
 $= \pi [\tan \frac{\pi}{3} - \tan 0] = \pi \sqrt{3}$   
 $= \pi \sqrt{3}$  cubic units (2m)



(i)  $\angle ADE = \angle CBE$  (alternate angles equal)  
 $AD \parallel BC$   
 and  $\angle DAE = \angle ECB$  (alternate angles equal)  
 $AD \parallel BC$   
 $\therefore \triangle ADE$  is isosceles triangle  
 $\therefore AE = DE$  (sides opposite equal angles are equal) (2m)

Question 13(d) continued:

(13)(d)(ii)  $\triangle BEC$  is isosceles

(since given  $\angle DBC = x^\circ = \angle ACB$ )

$\therefore EB = EC$  (sides opposite equal angles are equal)

But  $AC = AE + EC$

and  $DB = DE + EB$

but  $AE = DE$ , and  $EB = EC$

$\therefore AC = DB$ .

Now in  $\triangle ABC$ ,  $\triangle DCB$ :

$AC = DB$  (as shown) S

$\angle ACB = \angle DBC$  (given) A

$BC$  is common S

$\therefore \triangle ABC \cong \triangle DCB$  (SAS) (3m)

(13)(d)(iii)  $\angle ABC = \angle DCB$

(matching angles in congruent triangles are equal)

i.e.  $\angle ABD + \angle DBC = \angle DCA + \angle ACB$

But  $\angle DBC = x^\circ = \angle ACB$  (given)

$\therefore \angle ABD = \angle DCA$  (2m)

Question 14 (15 Marks)

(14)(a)(i)  $1 + (\sqrt{5}-2) + (\sqrt{5}-2)^2 + \dots$

Told this is geometric series:

have  $a=1$ ,  $r=\sqrt{5}-2$

Limiting sum exists if  $|r| < 1$

in this case  $r = \sqrt{5}-2 = 0.236\dots$

i.e.  $r < 1$

$\therefore$  limiting sum exists. (1m)

(14)(a)(ii)  $S_\infty = \frac{a}{1-r}$   
 $= \frac{1}{1-(\sqrt{5}-2)}$   
 $= \frac{1}{3-\sqrt{5}}$   
 $= \frac{1}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}}$   
 $= \frac{3+\sqrt{5}}{9-5}$   
 $\therefore S_\infty = \frac{3+\sqrt{5}}{4}$

Question 14 continued:

14)(b)  $x^2 + (k-2)x + 4 = 0$

$$\Delta = b^2 - 4ac$$

$$= (k-2)^2 - 4 \times 1 \times 4$$

$$= k^2 - 4k + 4 - 16$$

$$\Delta = k^2 - 4k - 12$$

$$\Delta = (k-6)(k+2)$$

for no real roots,

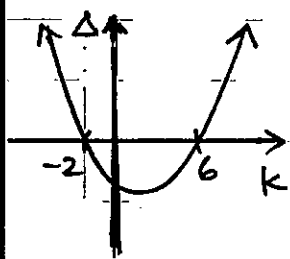
$$\Delta < 0$$

from sketch,

$$\Delta < 0 \text{ when}$$

$$-2 < k < 6$$

(2m)



14)(c)(i)  $x^2 + y^2 = 9$  is a circle centre (0,0) radius 3.

For  $3x - 4y = 15$  to be a tangent to circle,  $\perp$  distance from (0,0) to $3x - 4y - 15 = 0$  must be 3 units.

$$d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$$= \left| \frac{3 \times 0 - 4 \times 0 - 15}{\sqrt{3^2 + 4^2}} \right|$$

$$= \left| \frac{-15}{\sqrt{25}} \right|$$

$$= \frac{15}{5}$$

$$\therefore d = 3$$

(2m)

So perpendicular distance is 3 units

 $\therefore 3x - 4y = 15$  is tangent to circle  $x^2 + y^2 = 9$ 

14)(c)(ii)  $3x - 4y = 15$  crosses:

$$\text{x-axis: when } y=0: 3x=15$$

$$x=5$$

x-intercept is (5,0)

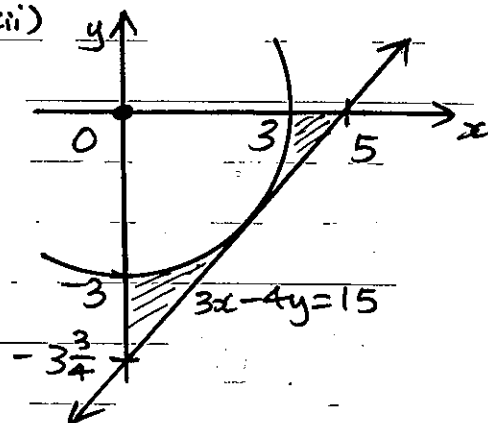
$$\text{y-axis: when } x=0: -4y=15$$

$$y = -\frac{15}{4}$$

$$y = -3\frac{3}{4}$$

Question 14(c)(ii) continued:

14)(c)(ii)



Shaded Area

$$= (\text{Area triangle}) - (\text{Area Quadrant})$$

$$= \left( \frac{1}{2} \times 5 \times \frac{15}{4} \right) - \left( \frac{1}{4} \times \pi \times 3^2 \right)$$

$$= \frac{75}{8} - \frac{9\pi}{4}$$

$$= \frac{75 - 18\pi}{8}$$

(2m)

$$= \frac{3(25 - 6\pi)}{8} \text{ square units}$$

14)(d)  $y = x^3 + x^2 - x + 2$

$$\frac{dy}{dx} = 3x^2 + 2x - 1$$

A and B are stationary points,

So need to solve  $\frac{dy}{dx} = 0$ :

$$3x^2 + 2x - 1 = 0$$

$$\begin{array}{r} x \quad +3 \\ -3 \quad -1 \\ \hline +2 \end{array}$$

$$3x^2 + 3x - x - 1 = 0$$

$$3x(x+1) - 1(x+1) = 0$$

$$(3x-1)(x+1) = 0$$

$$x = \frac{1}{3} \text{ or } x = -1$$

$$\text{when } x = -1: y = (-1)^3 + (-1)^2 - (-1) + 2$$

$$= -1 + 1 + 1 + 2$$

$$y = 3$$

$$\text{when } x = \frac{1}{3}: y = \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2 - \left(\frac{1}{3}\right) + 2$$

$$= \frac{49}{27}$$

$$y = 1\frac{22}{27}$$

$$\therefore A(-1, 3), B\left(\frac{1}{3}, 1\frac{22}{27}\right)$$

(3m)

Question 14 (d) continued:

$$(14)(d)(ii) \quad \frac{dy}{dx} = 3x^2 + 2x - 1$$

$$\frac{d^2y}{dx^2} = 6x + 2$$

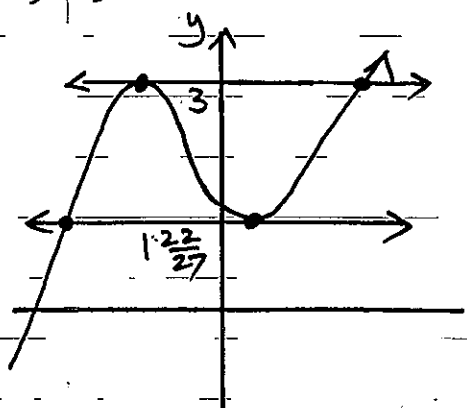
concave up when  $\frac{d^2y}{dx^2} > 0$

i.e. when  $6x + 2 > 0$

$$6x > -2$$

$\therefore$  concave up when  $x > -\frac{1}{3}$  (2m)

(14)(d)(iii)



$x^3 + x^2 - x + 2 = k$  represents the intersection of  $y = x^3 + x^2 - x + 2$  and  $y = k$ .

There will be 3 real solutions for

$$1 \frac{22}{27} < k < 3 \quad (1m)$$

(Note: only 2 roots if  $k = 1 \frac{22}{27}$  or  $k = 3$ )

Question 15 (15 Marks)

a) (i) After first card is drawn, 51 cards remain of which  $3 \times 13$  are of a different suit

$\therefore P(\text{2nd card different suit})$

$$= \frac{39}{51}$$

$$= \frac{13}{17} \quad (1m)$$

Question 15 (a) continued:

(15)(a)(ii)

$P(\text{all 4 different suits})$

$$= P(\text{card 1 any suit}) \times P(\text{card 2 any of 3 suits}) \times P(\text{card 3 any of 2 suits}) \times P(\text{card 4 last remaining suit})$$

$$= \left(\frac{52}{52}\right) \times \left(\frac{39}{51}\right) \times \left(\frac{26}{50}\right) \times \left(\frac{13}{49}\right)$$

$$= \frac{685464}{6497400}$$

$$= \frac{2197}{20825}$$

$$= 0.1054981993...$$

$$\doteq 0.105 \text{ (3 decimal places)} \quad (2m)$$

$$(15)(b)(i) \quad \ddot{x} = 4 \sin 2t$$

$$\dot{x} = \int \ddot{x} dt$$

$$= \int 4 \sin 2t$$

$$\dot{x} = -2 \cos 2t + C_1$$

when  $t=0$ ,  $\dot{x}=0$ :  $0 = -2 \cos 2(0) + C_1$

$$0 = -2 + C_1$$

$$\therefore C_1 = 2$$

$$\therefore \dot{x} = -2 \cos 2t + 2 \quad (2m)$$

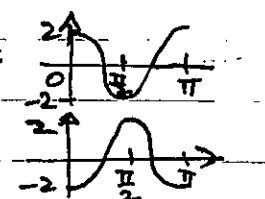
$$(15)(b)(ii) \quad \dot{x} = 2 - 2 \cos 2t$$

$$\text{Period} = \frac{2\pi}{2} = \pi$$

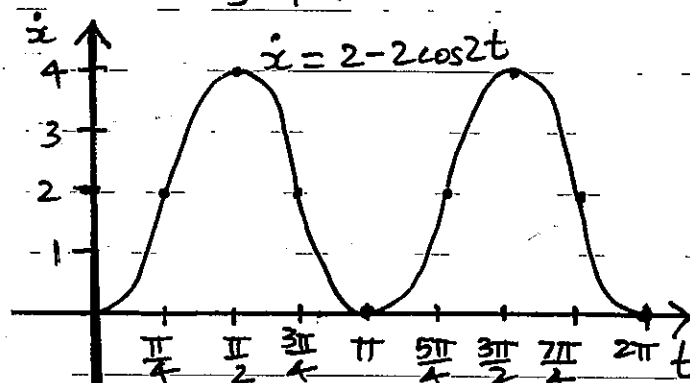
$$\text{Amplitude} = 2$$

$$\dot{x} = 2 \cos 2t$$

$$\dot{x} = -2 \cos 2t$$



then translate UP 2 units to get final sketch of graph:



Particle first comes to rest (after  $t=0$ )

$$\text{at } t = \pi \quad (3m)$$

Question 15(b) continued:

15)(b)(iii) Distance travelled between  
 $t=0$  and  $t=\pi$

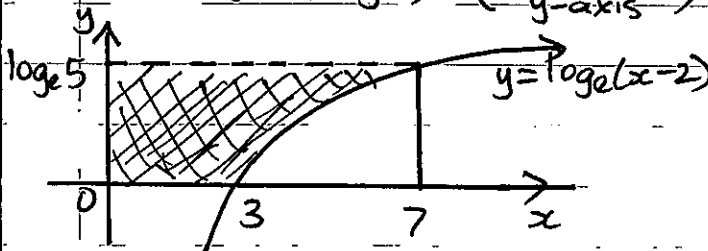
$$\begin{aligned} x &= \int_0^{\pi} (2 - 2\cos 2t) dt \\ &= [2t - \sin 2t]_0^{\pi} \\ &= (2\pi - \sin 2\pi) - (0 - \sin 0) \\ &= (2\pi - 0) - (0 - 0) \end{aligned}$$

Distance =  $2\pi$

$\therefore$  distance travelled is  $2\pi$  units (2m)

15)(c) Required area needs to be found by subtraction

Area = (area of rectangle) - (area between curve and y-axis)



Shaded area about y-axis needed:

$$y = \log_e(x-2)$$

From definition of logarithm:

$$e^y = x - 2$$

$$\therefore x = 2 + e^y$$

$$\begin{aligned} \text{Area} &= \int_0^{\log_e 5} (2 + e^y) dy \\ &= [2y + e^y]_0^{\log_e 5} \\ &= (2\log_e 5 + e^{\log_e 5}) - (2 \times 0 + e^0) \end{aligned}$$

$$= 2\log_e 5 + 5 - 1$$

$$\text{Area} = 2\log_e 5 + 4$$

Now Area Rectangle =  $7 \times \log_e 5$   
 $= 7\log_e 5$

So required Area

$$\begin{aligned} &= (\text{Area rectangle}) - (\text{area between curve \& y-axis}) \\ &= 7\log_e 5 - (2\log_e 5 + 4) \\ &= 5\log_e 5 - 4 \text{ square units} \end{aligned}$$

(5m)

Question 16 (15 Marks)

16)(a)(i)  $Q = Q_0 e^{-kt}$

when  $t=0$ ,  $Q=6$ :  $6 = Q_0 e^0$

$$\therefore Q_0 = 6$$

So  $Q = 6e^{-kt}$

when  $t=15$ ,  $Q = \frac{Q_0}{2}$

$$\therefore \frac{Q_0}{2} = Q_0 e^{-15k}$$

$$\frac{1}{2} = e^{-15k}$$

Taking logarithms base  $e$  of both sides:

$$\log_e\left(\frac{1}{2}\right) = -15k$$

$$k = -\frac{1}{15} \log_e\left(\frac{1}{2}\right)$$

$$= 0.04620981204\dots$$

$$\therefore k \approx 0.046 \text{ (3 decimal places)}$$

$\therefore Q_0 = 6$ ,  $k \approx 0.046$  (2m)

16)(a)(ii) Require  $\frac{1}{8}$  of initial dose  $Q_0$ :

i.e. require  $\frac{Q_0}{8}$

Solving  $\frac{Q_0}{8} = Q_0 e^{-kt}$

$$\frac{1}{8} = e^{-kt}$$

Taking logarithms base  $e$  of both sides:

$$\log_e\left(\frac{1}{8}\right) = -kt$$

$$t = \frac{\log_e\left(\frac{1}{8}\right)}{-k}$$

$$= \frac{\log_e\left(\frac{1}{8}\right)}{-0.04620981204}$$

$$= 45$$

$\therefore$  After 45 hours, there will be  $\frac{1}{8}$  of initial dose remaining

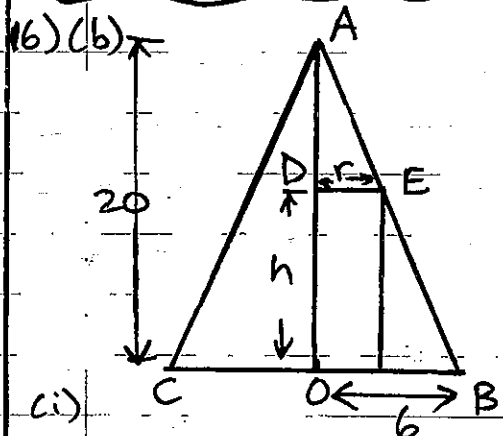
(2m)

ALTERNATIVE METHOD 16(a)(ii)

15 hours  $\rightarrow$  half life

$$\frac{1}{8} \rightarrow 3 \text{ "half lives"} \rightarrow \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

So  $\frac{1}{8}$  of initial dose =  $3 \times 15$   
 $= 45 \text{ hours}$

Question 16 continued:

(i)

Take cross-section of cone.

Since top of cylinder (DE) is parallel to base of cone (CB),

$$\triangle ADE \parallel \triangle AOB$$

(equiangular)

$$\therefore \frac{AD}{AO} = \frac{DE}{OB}$$

(matching sides similar triangles are in same ratio)

i.e.

$$\frac{20-h}{20} = \frac{r}{6}$$

$$6(20-h) = 20r$$

$$120 - 6h = 20r$$

$$6h = 120 - 20r$$

$$h = \frac{120 - 20r}{6}$$

$$h = \frac{20(6-r)}{6}$$

$$\therefore h = \frac{10(6-r)}{3}$$

 $\therefore$  Volume of cylinder

$$= \pi r^2 h$$

$$= \pi r^2 \cdot \frac{10(6-r)}{3}$$

$$\therefore V = \frac{10\pi r^2 (6-r)}{3} \quad (2m)$$

Question 16(b) continued:

$$16)(b) \text{ (ii)} \quad V = \frac{10\pi r^2 (6-r)}{3}$$

$$\text{i.e. } V = \frac{10\pi \cdot 6r^2}{3} - \frac{10\pi r^3}{3}$$

$$V = 20\pi r^2 - \frac{10\pi r^3}{3}$$

$$\therefore \frac{dV}{dr} = 40\pi r - \frac{3 \cdot 10\pi r^2}{3}$$

$$\frac{dV}{dr} = 40\pi r - 10\pi r^2$$

For stationary points, solve  $\frac{dV}{dr} = 0$ 

$$40\pi r - 10\pi r^2 = 0$$

$$10\pi r(4 - r) = 0$$

$$\therefore r = 0 \text{ or } r = 4$$

Require maximum volume, so  $r \neq 0$ Need to check  $r = 4$  is a maximum:

$$\frac{dV}{dr} = 40\pi r - 10\pi r^2$$

$$\frac{d^2V}{dr^2} = 40\pi - 20\pi r$$

$$\text{when } r = 4: \frac{d^2V}{dr^2} = 40\pi - 80\pi = -40\pi$$

$$\text{i.e. } \frac{d^2V}{dr^2} < 0 \text{ when } r = 4$$

i.e. concave down  $\searrow \searrow$  $\therefore$  maximum when  $r = 4$ .

$$\text{from (i): } h = \frac{10(6-r)}{3}$$

$$= \frac{10 \times 2}{3}$$

$$h = \frac{20}{3} \text{ or } 6\frac{2}{3}$$

 $\therefore$  Maximum volume when

$$r = 4 \text{ cm, } h = \frac{20}{3} \text{ cm or } 6\frac{2}{3} \text{ cm}$$

(2m)

Question 16 continued:

16) cc)  $A_n$  is amount owed at end of  $n$  months

$P$  is principal, amount borrowed.

in this case  $P = 80\,000$ .

(ci)  $A = P \left(1 + \frac{r}{100}\right)^n$

$$A_1 = 80\,000 \times \left(1 + \frac{2}{100}\right)^1$$

ie.  $A_1 = 80\,000 \times 1.02$

$$= \$81\,600 \quad (1m)$$

(cii)  $A_2 = A_1 \left(1 + \frac{2}{100}\right)$

$$= A_1 \times 1.02$$

$$= (80\,000 \times 1.02) \times 1.02$$

$$A_2 = 80\,000 \times 1.02^2$$

$$A_3 = A_2 \times 80\,000$$

$$= (80\,000 \times 1.02^2) \times 1.02$$

$$A_3 = 80\,000 \times 1.02^3$$

Following this pattern for first 12 months, then:

$$A_{12} = 80\,000 \times 1.02^{12}$$

So after 1st instalment of  $\$M$ ,

Amount owing after 12 months

$$= 80\,000 \times 1.02^{12} - M \quad (2m)$$

(cii) 1st repayment made after  $A_{12}$ .

But 2nd repayment not made until after  $A_{24}$ .

$$A_{13} = A_{12} \left(1 + \frac{2}{100}\right)$$

$$= A_{12} \times 1.02$$

$$A_{13} = [80\,000 \times 1.02^{12} - M] \times 1.02$$

$$A_{14} = [80\,000 \times 1.02^{12} - M] \times 1.02^2$$

$$A_{15} = [80\,000 \times 1.02^{12} - M] \times 1.02^3$$

Following this pattern:

$$A_{24} = [80\,000 \times 1.02^{12} - M] \times 1.02^{12}$$

Question 16(c)(cii) continued:

2nd (final) repayment of  $\$M$  is paid after  $A_{24}$ :

ie.  $[80\,000 \times 1.02^{12} - M] \times 1.02^{12} - M$

But after 2nd/final repayment there is no money owing.

$$\therefore [80\,000 \times 1.02^{12} - M] \times 1.02^{12} - M = 0$$

$$80\,000 \times 1.02^{24} - 1.02^{12}M - M = 0$$

$$80\,000 \times 1.02^{24} - M(1.02^{12} + 1) = 0$$

ie.  $M(1.02^{12} + 1) = 80\,000 \times 1.02^{24}$

$$\therefore M = \frac{80\,000 \times 1.02^{24}}{(1.02^{12} + 1)}$$

ie.  $M = \frac{80\,000 \times (1.02)^{24}}{(1.02^{12}) + 1} \quad (2m)$

(iv) From calculator

$$M = \frac{80\,000 \times 1.02^{24}}{1.02^{12} + 1} = 56\,728.95203$$

ie. each instalment  $M = \$56\,728.95$

$\therefore$  Total repaid in the 2 instalments

$$= 2 \times \$56\,728.95$$

$$= \$113\,457.90$$

$$\therefore \text{Interest} = (\text{Total repaid}) - (\text{original loan of } \$80\,000)$$

$$= \$113\,457.90 - \$80\,000$$

$$= \$33\,457.90$$